

Bureau of Indian Standards

Manak Bhawan, 9, Bahadur Shah Zafar Marg
New Delhi - 110002

IS 16001:2012

2012

FLY ASH FOR CEMENT MANUFACTURE

SPECIFICATION

ICS 91.100.10

BIS 2012

Indian Standard

1 SCOPE

This standard specifies the requirements for fly ash generated from burning bituminous or sub-bituminous coal in thermal power plants, for use as a raw material in the manufacture of Portland cement.

2 DEFINITIONS

2.1 Fly Ash

The finely divided residue resulting from the combustion of pulverized coal in thermal power plants, collected by means of electrostatic precipitators or mechanical devices.

3 CHEMICAL COMPOSITION

Oxide	Class F (Bituminous)	Class C (Sub-bituminous)
SiO ₂ + Al ₂ O ₃ + Fe ₂ O ₃ , %, min	70.0	50.0
SO ₃ , %, max	3.0	5.0
CaO, %, max	10.0	25.0
MgO, %, max	5.0	5.0
Loss on Ignition, %, max	5.0	5.0
Moisture Content, %, max	1.0	1.0

4 PHYSICAL PROPERTIES

Property	Requirement
Fineness (45 m sieve), % retained, max	35
Soundness (Le Chatelier), mm, max	5
Compressive Strength Ratio (28 day), min	0.80
Activity Index (7 day), min	65% of control

5 TEST METHODS

- Fineness: IS 4031 (Part 1) - Wet Sieve Method
- Chemical Analysis: IS 13821 - X-ray Fluorescence
- Compressive Strength: IS 4031 (Part 6) - Mortar Cubes
- Loss on Ignition: IS 13821 - Furnace at 950C